

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Forth Semester of B. Tech. (Civil) Examination
May 2011

CL 207 Concrete Technology

Date: 09.05.2010,

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) State the advantages and disadvantages of concrete. 04
- (b) What is the role of water in quality of concrete? List three impurities in water and its impact on concrete. 03
- Q - 2 (a) Discuss briefly with diagrammatic sketch dry process of manufacture of Portland Cement. 03
- (b) Explain the process of hydration of cement 04
- (c) What is a Fly Ash? State different types of fly Ash. Explain briefly its effect on fresh concrete. 03
- (d) Discuss Sulphate Resisting Cement and it's uses 04

OR

- Q - 2 (a) State the chemical composition of Portland Cement, and explain the role of each compound. 03
- (b) Discuss with neat sketch the composition of cement paste at different stages of hydration. 04
- (c) How ground granulated blast furnace slag is obtained? Explain its effect on hardened concrete. 03
- (d) How is Portland Pozzolana Cement different from Ordinary Portland cement? State two advantages and two application of PPC. 04
- Q - 3 (a) What is the role of aggregate in concrete? Explain the bulking of aggregate. How will you make necessary correction of bulking in concrete mix design? 04
- (b) Explain briefly aggregate crushing value test in laboratory with its significance 03
- (b) What are admixtures? Explain briefly its uses. 07
- What is a plasticizer? Explain briefly its effect in concrete.

OR

- Q - 3 (a) What is alkali aggregate reaction? Explain the factors affect it. 04
- (b) Explain the sieve Analysis test performed on fine aggregate in laboratory. 03
- (c) State advantages of admixtures. Explain briefly difference between retarders and accelerators in relation to its properties and use. 07

SECTION - II

- Q - 4 (a)** Design a concrete mix as per IS : 10262, I.S. method for following data : 07
- Characteristic strength at 28 days = 25MPa
Strength of cement at 28 day = 51 N/mm²
Maximum size of aggregate = 20 mm
Degree of workability = medium, Compaction factor = 0.9
Quality control = Good
Type of exposure = Severe
Specific gravity of
- (i) Cement = 3.15
 - (ii) Coarse aggregate = 2.80
 - (iii) Fine aggregate = 2.60
- It is found that sand falls in grade zone III. Mix is designed in such a way that only one sample fails out of 20. The amount of entrapped air is 2%. For require data refer annex 1.
Find proportion of water, Fine Aggregate and Coarse aggregate for one bag of cement.
- Q - 5 (a)** State significance of size of aggregate in mix design. 03
- (b)** State workability and effect of workability on properties of fresh concrete. 06
- (c)** State requirements and properties of high performance concrete. 05
- OR**
- Q - 5 (a)** State the factors governing the selection of mix proportion. 03
- (b)** Write process of manufacturing of fresh concrete. 06
- (c)** Write short note on fiber reinforced concrete. 05
- Q - 6 (a)** Explain what you understand by segregation and bleeding of fresh concrete. Also state the preventive measure to control it. 04
- (b)** Describe the importance of curing? When should it be start? Describe various technique of curing. 05
- (c)** State various methods to determine compressive strength of concrete. 05
- OR**
- Q - 6 (a)** Describe compaction factor test to determine workability of concrete. 04
- (b)** How porosity affects the durability and strength of concrete 05
- (c)** Explain and draw the relation of water cement ratio and methods of compaction on compressive strength of concrete. 05

TABLE 1 SUGGESTED VALUES OF STANDARD DEVIATION

GRADE OF CONCRETE	STANDARD DEVIATION FOR DIFFERENT DEGREE OF CONTROL IN N/mm ²		
	Very Good	Good	Fair
(1)	(2)	(3)	(4)
M 10	2.0	2.3	3.3
M 15	2.5	3.5	4.5
M 20	3.6	4.6	5.6
M 25	4.3	5.3	6.3
M 30	5.0	6.0	7.0
M 35	5.3	6.3	7.3
M 40	5.6	6.6	7.6
M 45	6.0	7.0	8.0
M 50	6.4	7.4	8.4
M 55	6.7	7.7	8.7
M 60-	6.8	7.8	8.8

NOTE — Appendix A provides guidance regarding the different degrees of quality control to be expected, depending upon the infrastructure and practices adopted at the construction site.

TABLE 2 VALUES OF :

(Clause 2.2)

**ACCEPTED PROPORTION
OF LOW RESULTS**

1 in 5	0.84
1 in 10	1.28
1 in 15	1.50
1 in 20	1.65
1 in 40	1.86
1 in 100	2.33

TABLE 3 APPROXIMATE AIR CONTENT

NOMINAL MAXIMUM SIZE OF AGGREGATE mm	ENTRAINED AIR, AS PERCENTAGE OF VOLUME OF CONCRETE
10	3.0
20	2.0
40	1.0

TABLE 4 APPROXIMATE SAND AND WATER CONTENTS PER CUBIC METRE OF CONCRETE FOR GRADES UPTO M 35

NOMINAL MAXIMUM SIZE OF AGGREGATE mm	WATER CONTENT*, PER CUBIC METRE OF CONCRETE kg	SAND AS PERCENT OF TOTAL AGGREGATE BY ABSOLUTE VOLUME
10	208	40
20	186	35
40	165	30

*Water content corresponding to saturated surface dry aggregate.

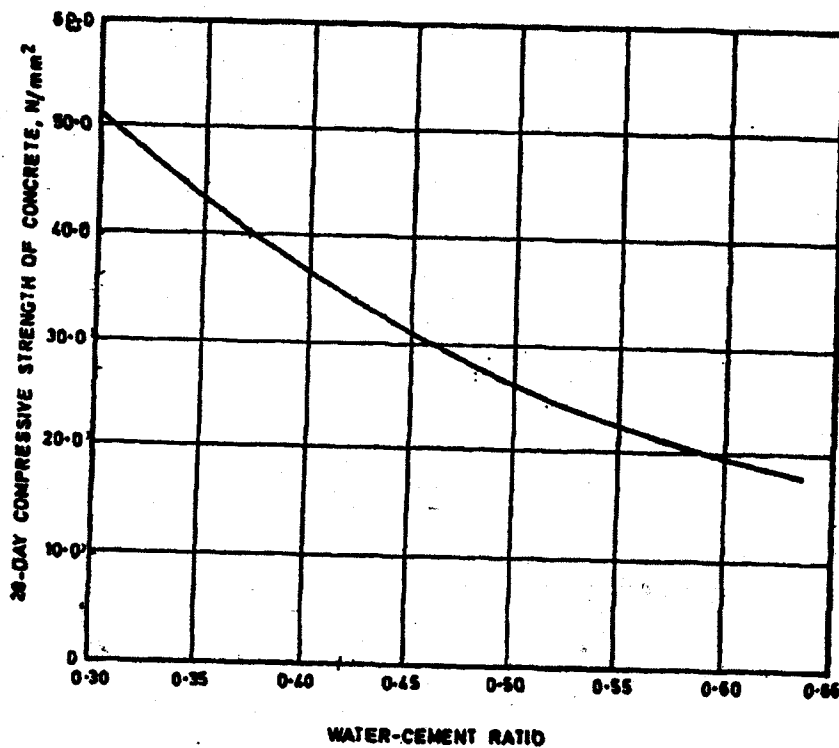


FIG. 1 GENERALISED RELATION BETWEEN FREE WATER-CEMENT RATIO AND COMPRESSIVE STRENGTH OF CONCRETE

TABLE 5 APPROXIMATE SAND AND WATER CONTENTS PER CUBIC METRE OF CONCRETE FOR GRADES ABOVE M 35

(Clauses 3.3.1, 3.3.3, 3.3.4 and Table 6)

NOMINAL MAXIMUM SIZE OF AGGREGATE mm	WATER CONTENT*, PER CUBIC METRE OF CONCRETE kg	SAND AS PERCENT OF TOTAL AGGREGATE BY ABSOLUTE VOLUME
10	200	28
20	180	25

*Water content corresponding to saturated surface dry aggregate.

TABLE 6 ADJUSTMENT OF VALUES IN WATER CONTENT AND SAND PERCENTAGE FOR OTHER CONDITIONS

(Clauses 3.3.4 and 4.1)

CHANGE IN CONDITION STIPULATED FOR TABLES 4 AND 5	ADJUSTMENT REQUIRED IN	
	Water Content	Percent, Sand in Total Aggregate
(1)	(2)	(3)
For sand conforming to grading Zone I, Zone III or Zone IV of Table 4 of IS : 383-1970*	0	+ 1.5 percent for Zone I - 1.5 percent for Zone III - 3.0 percent for Zone IV
Increase or decrease in the value of compacting factor by 0.1	± 3 percent	0
Each 0.05 increase or decrease in free water-cement ratio	0	± 1 percent
For rounded aggregate	- 15 kg/m ³	- 7 percent

*Specification for coarse and fine aggregates from natural sources for concrete (second revision).

IS : 456 - 2000

Table 5 Minimum Cement Content, Maximum Water-Cement Ratio and Minimum Grade of Concrete for Different Exposures with Normal Weight Aggregates of 20 mm Nominal Maximum Size

(Clauses 6.1.2, 8.2.4.1 and 9.1.2)

Sl No.	Exposure	Plain Concrete			Reinforced Concrete		
		Minimum Cement Content kg/m ³	Maximum Free Water-Cement Ratio	Minimum Grade of Concrete	Minimum Cement Content kg/m ³	Maximum Free Water-Cement Ratio	Minimum Grade of Concrete
I)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Mild	220	0.60	-	300	0.55	M 20
iii)	Moderate	240	0.60	M 15	300	0.50	M 25
iii)	Severe	250	0.50	M 20	320	0.45	M 30
iv)	Very severe	260	0.45	M 20	340	0.45	M 35
v)	Extreme	280	0.40	M 25	360	0.40	M 40